TERESA: Whole-Body Active Perception for Legged Robot-Assisted Emergency Assessment

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***Abstract*—Deploying legged mobile manipulators for autonomous emergency assessment could extend care to inaccessible environments, but requires the robot to actively reconfigure its body to compensate for perceptual uncertainty while approaching an unknown patient.**

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA), an autonomous emergency assessment system on a Boston Dynamics Spot quadruped with a Unitree Z1 arm. TERESA introduces a whole-body active perception framework that continuously adapts body posture—height, pitch, and yaw—to maximise perceptual quality. A hybrid confidence-gated search combines coarse body rotation with targeted refinement, confirming patient localisation through a dual-sensor lock before navigation. An adaptive anticipatory scanner acquires multi-view anatomical data concurrently with base motion, dynamically reducing scanning poses when torso keypoints are already resolved, eliminating idle time of sequential approaches. An exposure body scanning pipeline covers the patient's body surface across all postures, integrating pre-trained forensic wound and burn detection models with an interactive review phase in which a single 2D click on the web camera feed triggers a full-body reconfiguration—posture optimisation, base navigation, and arm trajectory replay—to reach the corresponding 3D body surface point. During Focused Assessment with Sonography in Trauma (FAST) scanning, pre-planned body reconfiguration optimises posture for each anatomical target via offline search, maximising arm manipulability without real-time whole-body optimisation.

Experiments on the physical Spot+Z1 platform demonstrate fully autonomous emergency assessment on supine patients, with the active perception pipeline reducing idle time compared to a stop-and-manipulate baseline. To the best of our knowledge, TERESA is the first legged system to demonstrate perception-driven whole-body reconfiguration with an interactive operator review interface for autonomous emergency assessment.

***Index Terms*—Whole-body active perception, legged manipulation, medical robotics, emergency assessment, mobile manipulation, body reconfiguration.**

I. Introduction

EMERGENCY assessment protocols—including the Airway, Breathing, Circulation, Disability, Exposure (Airway, Breathing, Circulation, Disability, Exposure (ABCDE)) survey and Focused Assessment with Sonography in Trauma (FAST) ultrasound [1]—enable rapid triage of life-threatening conditions at the point of care. In mass-casualty incidents and disaster sites, however, trained personnel are often unavailable, creating a critical bottleneck. A legged mobile manipulator could autonomously navigate unstructured terrain to locate patients and perform these assessments, but faces a fundamental challenge: the robot must actively reconfigure its body to compensate for perceptual uncertainty, as the quality of patient detection depends critically on how it positions its own camera—viewpoint, height, and orientation all affect accuracy. This coupling between body posture and perceptual quality has no closed-form model, and in emergency scenarios the robot operates with only onboard sensing.

Robot-assisted ultrasound has advanced over two decades on fixed-base platforms. Comprehensive surveys [2]–[5] document the evolution from teleoperated systems to contemporary autonomous scanning, and Hidalgo et al. [6] review current clinical applications. Key technical advances include force/position decoupling [7], impedance control on curved surfaces [8], force-guided scanning for soft tissue [9], active-sensing end-effectors [10], and autonomous protocol execution [11]–[13]. Building on assistive robotics, Gu and Demiris [14] proposed visuo-tactile learning for robot-assisted bed bathing, and Sayols et al. [15] developed a multi-layer motion planner for autonomous surgical robots. Despite this progress, all deployed systems share a fundamental limitation: the patient must be pre-positioned within the arm's reachable workspace, confining autonomous ultrasound to controlled clinical settings.

For the Exposure component of ABCDE—the systematic visual inspection of the body surface for wounds, burns, and haemorrhages—computer vision models have recently emerged. Pre-trained YOLOv8-based forensic wound detectors[1] identify lacerations and contusions, while YOLOv7-based burn classifiers[2] grade burn severity across three degrees. These models have demonstrated effective classification on clinical datasets but have not been deployed on mobile robotic platforms. For 3D body pose estimation, Neural Localizer Fields (NLF) [16] regress 24-joint SMPL skeletons directly from monocular RGB, providing full anatomical coverage including spine, hands, and feet absent from the 17-keypoint COCO format. Addressing uncertainty in human-centered robotics, Casado et al. [17] proposed diffusion-based intention estimation for shared wheelchair control—a theme that similarly motivates TERESA's quality-driven perception loop.

Legged platforms such as the Boston Dynamics Spot remove the workspace constraint—they can traverse stairs, rubble, and confined spaces inaccessible to wheeled robots, and can therefore reach patients wherever they lie. Mounting a

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[1]Available at https://huggingface.co/trapezius60/forensic_wound_detection

[2]Available at https://github.com/Michael-OvO/Skin-Burn-Detection-Classification

manipulator on a legged base introduces, however, a deeper challenge. Operational space formulation [18], manipulability analysis [19], [20], and optimisation-based control [21] underpin modern Whole-Body Control (WBC). For legged manipulators, hierarchical Model Predictive Control (MPC) [22], [23], holistic [24] and Quadratic Program (QP)-based [25] WBC, visual feedback integration [26], and Spot-specific grasping [27] and manipulability enhancement [28] have been demonstrated. All existing legged WBC formulations, however, assume known or static manipulation targets and do not account for the coupling between body posture and perceptual quality.

Active perception surveys [29], [30] establish that moving to improve sensing outperforms passive observation [31]. Perception-aware planning [32], [33], next-best-view grasping [34], view planning for pose estimation [35], and active neural sensing [36] couple motion with perception quality. For mobile manipulation, whole-body interaction [37], active-perceptive motion [38], coupled perception-manipulation [39], and one-shot multimodal frameworks [40] optimise trajectories for information gain, yet operate on wheeled bases whose body posture is fixed. Posture optimisation via reachability [41], [42], base placement [43], body-ground contact [44], and legged manipulation surveys [45] treat perception as input rather than an optimisation objective. No existing system closes the loop between body posture and perceptual quality on a legged platform for medical tasks.

We advocate a paradigm in which body posture on a legged platform is treated as a first-class perceptual resource: the robot actively adapts its height, pitch, and yaw to maximise the quality of its own perceptual input, rather than solely to minimise kinematic error. The key idea is to close the perception-action loop at the whole-body level through confidence-gated search and anticipatory scanning. This paper addresses three fundamental questions: (i) how can a legged robot reconfigure its body to maximise perceptual quality when the target position is unknown, (ii) how can anatomical scanning be performed concurrently with base navigation to eliminate the idle time of sequential approaches, and (iii) how can body posture be pre-planned to maximise arm manipulability across multiple scanning targets without requiring real-time whole-body optimisation?

This paper presents Trustworthy Emergency Robot for Efficient Support and Assistance (TERESA), an integrated system for autonomous emergency assessment on a Boston Dynamics Spot legged robot equipped with a Unitree Z1 manipulator arm. Building on the fixed-base ultrasound scanning competence established in [13], TERESA extends autonomous assessment to a fully mobile platform through a whole-body active perception framework. The system integrates the pre-trained wound and burn detection models into its onboard perception pipeline, running inference on the End-Effector (EE)-mounted RealSense camera during the exposure body scan, and provides an interactive web-based review interface enabling operator-guided re-inspection of detected injuries.

The key contributions of this paper are as follows.

1) **Whole-body active perception framework** in which body posture—height, pitch, and yaw—is optimised to improve perceptual quality rather than solely to minimise kinematic error (Section III).
2) **Hybrid confidence-gated search strategy** that combines coarse body rotation with targeted pitch refinement and dual-sensor locking, ensuring robust patient localisation before the robot commits to navigation.
3) **Exposure body scanning pipeline** that acquires multi-view anatomical data through an adaptive skeleton-keypoint grid, integrating pre-trained forensic wound and burn classification models with an interactive web-based review interface (Sections III and IV).
4) **Pre-planned body reconfiguration method** that optimises the legged base posture for each anatomical scanning target through offline grid search, maximising arm manipulability without real-time whole-body optimisation.
5) **Experimental validation** on a physical Spot+Z1 platform demonstrating fully autonomous emergency assessment with reduced idle time compared to a sequential stop-and-manipulate baseline (Section V).

To the best of our knowledge, TERESA is the first legged robotic system to demonstrate perception-driven whole-body reconfiguration for autonomous emergency assessment.

The remainder of this paper is organised as follows. Section II defines the system and the autonomous emergency assessment problem. Section III presents the whole-body active perception and control framework, including search, scanning, reconfiguration, and exposure body scanning with interactive review. Section IV details the system architecture, mission Finite State Machine (FSM), and web-based operator interface. Section V presents experiments and results. Section VI concludes the paper.

II. Problem Statement

A. System Model

The TERESA platform consists of a Boston Dynamics Spot quadruped and a Unitree Z1 six-Degree of Freedom (DOF) arm rigidly mounted on the robot's back. Four frames are defined: the world frame $\mathcal{F}_W$ (coincident with Spot odometry), the body frame $\mathcal{F}_B$, the arm base frame $\mathcal{F}_0$ (fixed relative to $\mathcal{F}_B$), and the end-effector frame $\mathcal{F}_{EE}$ (at the Z1 flange). The Spot base is modelled as a non-holonomic vehicle with configuration $\boldsymbol{\xi} = [x,\, y,\, \theta]^\top \in \mathbb{R}^2 \times \mathbb{S}^1$, constrained to forward velocity v_x and yaw rate ω_z:

$$\dot{\boldsymbol{\xi}} = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix} v_x + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \omega_z. \tag{1}$$

The body posture is parametrised by height h and pitch φ, commanded through the native `body_pose` interface, with bounds $h \in [h_{\min},\, h_{\max}]$, $\varphi \in [0,\, \varphi_{\max}]$. The Z1 arm has $n = 6$ revolute joints $\mathbf{q} \in \mathbb{R}^6$ with geometric Jacobian $\mathbf{J}_{\text{arm}}(\mathbf{q}) \in \mathbb{R}^{6\times 6}$. The combined control input during the approach phase is $\mathbf{u} = [\dot{\mathbf{q}},\, v_x,\, \omega_z]^\top \in \mathbb{R}^8$.

B. Task Definition

An autonomous emergency assessment mission requires the EE to reach a sequence of N anatomical target poses $\mathcal{T} = \{\mathbf{T}_i^* \in SE(3)\}_{i=1}^N$ (e.g., FAST quadrants). The targets are not known a priori—they are estimated during the mission from the onboard RGB-D (RGBD) cameras, which are the only source of perception.

C. Problem

Given the system model and the unknown patient position, find control inputs $\mathbf{u}(t)$ and body posture commands $(h(t), \varphi(t))$ that enable the robot to: (i) discover a posture from which detection confidence exceeds a threshold, ensuring robust patient localisation before committing to the approach; (ii) navigate to a configuration from which all targets in $\mathcal{T}$ are reachable; (iii) reconfigure its body to maximise arm manipulability for each scanning target; and (iv) bring the EE to each target pose with bounded error $\|\mathbf{p}_{EE} - \mathbf{p}_i^*\| \le \varepsilon_p$.

D. Assumptions

The following assumptions are adopted: the patient remains stationary during scanning (A1); Spot operates on flat terrain (A2); the rigid transform between $\mathcal{F}_B$ and $\mathcal{F}_0$ is known and constant (A3); a single patient is present (A4); and no external perception infrastructure is available (A5).

III. Whole-Body Active Perception and Control

The core contribution of TERESA is a whole-body active perception framework: the robot continuously adapts its posture—body height, pitch, and yaw—to maximise perceptual quality. This section describes the three mechanisms that realise this principle: hybrid confidence-gated active search (Section III-A), adaptive anticipatory scanning with WBC (Section III-B), and pre-planned body reconfiguration with compliant contact (Section III-C).

A. Hybrid Confidence-Gated Active Search

Before the robot can approach the patient, it must first localise them reliably. In an unstructured emergency scenario, the patient's position is unknown a priori, and the detection quality of the onboard Orbbec RGBD camera depends strongly on the robot's body posture: a higher camera provides a wider field of view but reduces resolution at the ground plane; a forward pitch directs the camera toward the expected patient location but narrows the horizon. TERESA treats this coupling as an opportunity rather than a limitation: the robot actively searches over body postures to find the configuration that maximises detection confidence.

1) Coarse rotation with targeted refinement: The robot sweeps through a sequence of body yaw orientations using velocity control, pausing at each to allow two cameras to observe the environment. A wide-field Orbbec RGBD camera mounted on Spot's body runs You Only Look Once (YOLO)11-pose [46] and a posture classifier that outputs a confidence score $c \in [0, 1]$. An EE-mounted RealSense RGBD camera runs a torso tracker in a lenient guidance mode. At each orientation, two conditions are evaluated:

- **Orbbec direct detection**: if the wide-field camera detects a lying patient with high confidence, the robot transitions immediately to target locking.
- **RealSense guidance**: if the EE camera detects a potential body while the Orbbec has not yet confirmed, the robot enters a semi-locking phase. It rotates the base toward the detected torso and tilts forward, giving the Orbbec a clean observation window to confirm.

If neither condition is met after the coarse sweep, the robot triggers a **refinement phase**: at the most promising orientation, it performs a targeted sweep of body pitch angles, maximising the Orbbec confidence score. This two-tier strategy combines the speed of a sparse coarse search with the precision of targeted refinement.

2) Dual-sensor confidence lock: Patient localisation is confirmed through a hybrid lock that fuses information from both cameras. When the Orbbec detects a lying posture with high confidence, the robot enters a locking state: the arm returns to an elevated home pose, and the coordinator collects a sequence of approach point estimates in the world-fixed odometry frame $\mathcal{F}_W$, computing their mean as the navigation target. Concurrently, the NLF pipeline produces a refined 3D skeleton estimate by accumulating two valid detections with exponential moving average smoothing, providing higher-fidelity anatomical data for the subsequent approach phase. The transition to the approach phase is gated by both the arm's home trajectory and the collection of the required number of samples. If the Orbbec detection is temporarily lost, the system tolerates a brief absence before resuming the search from the current orientation.

B. Adaptive Anticipatory Scanning with Whole-Body Control

In traditional mobile manipulation pipelines, the robot first navigates to a target position, stops, and only then begins visual scanning—a sequential approach that introduces significant idle time. TERESA eliminates this idle period by executing the body scan concurrently with the approach navigation.

1) Whole-body coordination: During the approach, two independent controllers run in parallel. The Spot base is driven by a proportional controller that computes v_x and ω_z from the distance and angle to the target in $\mathcal{F}_W$. The arm is controlled by a WBC formulation that orients the EE-mounted RealSense toward the patient's torso centroid. The joint velocities are computed via a damped pseudo-inverse of the arm Jacobian:

$$\dot{\mathbf{q}} = \mathbf{J}_{\text{arm}}^\top \left(\mathbf{J}_{\text{arm}} \mathbf{J}_{\text{arm}}^\top + \mu \, \mathbf{I}_6\right)^{-1} \mathbf{v}_{\text{des}}, \tag{2}$$

where $\mu > 0$ is a damping factor and $\mathbf{v}_{\text{des}} \in \mathbb{R}^6$ encodes the desired angular velocity toward the target, computed from the orientation error between the current EE orientation and the desired look-at direction $\mathbf{x}_{EE} = (\mathbf{p}^* - \mathbf{p}_{EE})/\|\mathbf{p}^* - \mathbf{p}_{EE}\|$. The translational component is zero, as the arm maintains a nominal home position during approach and contributes only orientation. The base velocity is modulated by a QualityMonitor that scales the approach speed in proportion to detection confidence, avoiding premature commitment to uncertain targets while preventing unnecessary stops.

2) Adaptive Cartesian scan grid: The arm executes a perceptual body scan using a dynamically generated Cartesian grid of poses around the current EE position. During the pre-approach phase, the RealSense tracker publishes per-keypoint confidence scores for the four torso landmarks (shoulders and hips), which drive the grid generation. If all four keypoints are detected with high confidence, the scan uses a minimal two-pose grid: the current EE pose advanced toward the patient plus an elevated viewpoint. If any keypoint falls below the threshold, the grid expands to a three-pose comprehensive pattern with symmetric lateral offsets, an elevated frontal viewpoint, and the home pose interleaved between lateral transits. All scanning poses include a forward advance toward the patient. At each pose, depth and skeleton data are fused into 3D keypoint estimates via a weighted average accounting for detection rate and spatial stability. From the fused torso model, the five canonical FAST target poses are computed as fixed offsets from the estimated torso centroid in $\mathcal{F}_0$.

C. Pre-Planned Body Reconfiguration and Contact Control

1) Posture optimisation: Once the anticipatory body scanner has localised the patient's anatomy and computed the target set $\mathcal{T}$ (both FAST ultrasound targets and exposure grid points) in $\mathcal{F}_W$, the system pre-plans an optimal body posture for each target through a two-tier grid search with Inverse Kinematics (IK)-driven retry. The optimisation is shared by both the FAST scanning and exposure scanning pipelines.

The posture optimisation space spans three parameters: body height $h \in [h_{\min}, h_{\max}]$ and pitch $\varphi \in [0, \varphi_{\max}]$, as defined in Section II, plus a body-frame base displacement $dy \in \mathbb{R}$ along the patient's head-to-foot axis, expressed in the `patient_body` frame (Section IV). The discrete search grids are $\mathcal{H} = \{h_1, \ldots, h_{N_h}\}$ and $\mathcal{P} = \{\varphi_1, \ldots, \varphi_{N_p}\}$, with dy sampled uniformly at $0.10\,\mathrm{m}$ intervals. The cost function includes a regularisation term that penalises lateral displacement:

$$J(h, \varphi, dy) = \left\| \mathbf{T}_{\mathcal{F}_W}^{\mathcal{F}_0}(h, \varphi, dy)\, \mathbf{p}^* - \mathbf{p}_{\text{sweet}} \right\| + \lambda\, |dy|, \quad (3)$$

where $\mathbf{p}_{\text{sweet}} \in \mathbb{R}^3$ is the empirically determined centre of the Z1 arm's preferred manipulation region in $\mathcal{F}_0$, and $\lambda = 0.50$ weights lateral movement against reachability. The value $\lambda = 0.50$ was chosen empirically to balance reachability against energy cost: $\lambda > 1.0$ prevented necessary base displacement, while $\lambda < 0.25$ caused excessive walking for marginal reachability gains. This penalty encourages the optimiser to prefer height and pitch adjustments over base walking—reducing energy consumption and producing smoother, more predictable motion—while still allowing longitudinal displacement when strictly necessary. The two-tier escalation proceeds as follows:

1) **2D search**—(h, φ) only: a coarse grid sweeps $\mathcal{H} \times \mathcal{P}$ ($N_h N_p$ combinations, typically 12). For each candidate, the system transforms the target to $\mathcal{F}_0$ and queries the IK solver. If a posture yields a solution within a $2\,\mathrm{s}$ timeout, it is selected.
2) **3D escalation**—(dy, h, φ): when the 2D search fails, the optimiser adds longitudinal base displacement dy (Spot walks along the patient's body), expanding to $N_{dy} N_h N_p$ combinations. If the IK solver also times out at this tier, the point is skipped—the arm cannot reach it even with full base repositioning.

The optimisation is computed once per target after $\mathcal{T}$ is available, imposing no real-time computational burden. Body reconfiguration is executed between targets: Spot assumes the pre-computed posture (h_i^*, φ_i^*), settles, and the base navigator drives the dy displacement before the Z1 arm executes the IK trajectory toward the transformed target.

2) Contact control: The Unitree Z1 does not provide native impedance control, yet ultrasound scanning requires compliant probe contact. TERESA implements a custom Cartesian impedance controller on the Z1 torque interface at $500\,\mathrm{Hz}$, computing the desired EE position relative to the detected torso surface centroid and outward normal. Joint torques are computed via the Jacobian transpose with adaptive stiffness and damping gains that scale with arm extension, with gravity and Coriolis compensation via Pinocchio. A safe switching service alternates with a joint trajectory controller for position-controlled motions, defaulting to position control on error.

D. Exposure Body Scanning and Injury Detection

After the anticipatory scanner has localised the patient, TERESA executes an exposure body scan that systematically covers the patient's body surface with the EE-mounted RealSense camera. The scan serves the "E" (Exposure) component of the ABCDE protocol, searching for visible wounds, burns, and haemorrhages. A key design principle is that the operator remains in the loop: the system automates the scanning motion but preserves the clinician's ability to inspect, re-examine, and override.

1) Posture-adaptive scan grid: The scan grid adapts to the patient's posture, generating a top-down bilinear grid over the torso interpolated between shoulder and hip keypoints. When the NLF prior is available, all 24 joints are populated, enabling a denser grid.

2) Injury detection: At each grid point, two pre-trained models run on the RealSense RGB stream: a YOLOv8-based forensic wound detector and a YOLOv7-based burn classifier at $5\,\mathrm{Hz}$ in alternating frames. Detections are projected to 3D world coordinates via the registered depth channel and associated with their grid point index.

3) Web-based interactive review: Following the sweep, the system enters an interactive review phase. The web interface displays the exposure grid overlaid on the RealSense feed, with detected injuries highlighted as colour-coded bounding boxes. The operator can click any grid point to command the physical robot to return to that body region: Spot reconfigures to the previously optimised posture, the arm replays the saved IK trajectory, and the camera returns to the original observation pose. This click-to-revisit mechanism runs through rosbridge WebSocket with sub-second latency. The operator freely navigates between points by clicking; a termination command ends the review and advances the mission to the FAST ultrasound sequence.

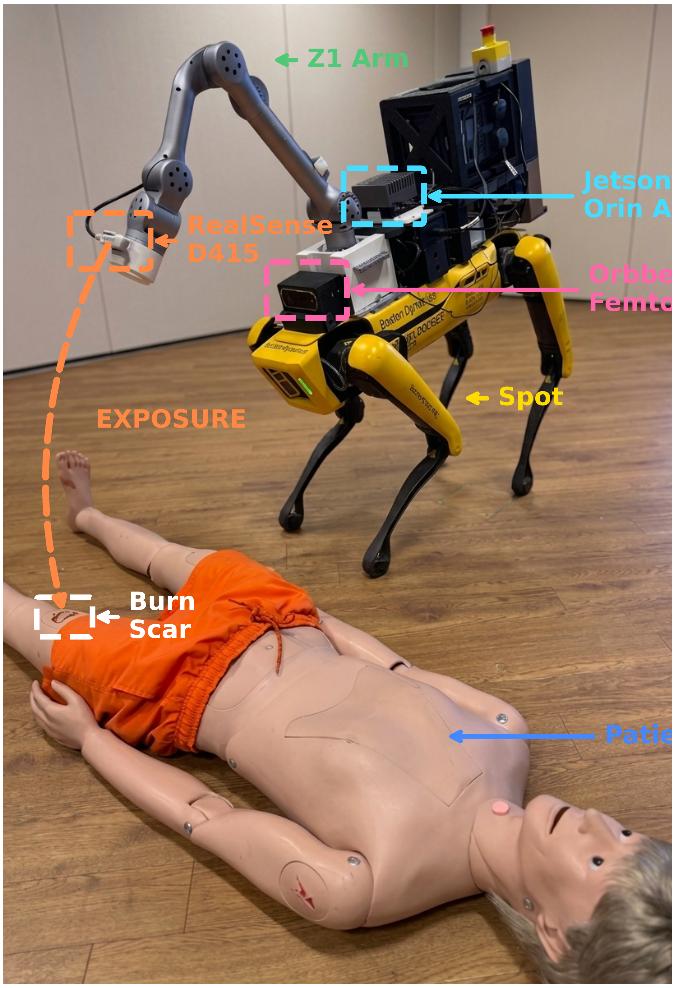

Fig. 1. The TERESA system during the exposure scanning phase. The Spot quadruped base (left) carries the Z1 manipulator with a wrist-mounted Intel RealSense D415 for close-range torso tracking and a body-mounted Orbbec Femto Bolt for wide-field patient detection. An NVIDIA Jetson Orin AGX provides on-board perception and arm control. The web-based Body Map interface (background) renders the real-time skeleton, body region boundaries, and exposure scan grid. The dashed orange arrow marks the exposure path from the RealSense to a burn scar on the patient's torso.

IV. System Architecture

The TERESA system is organised into four layers—hardware, perception, mission management, and control—communicating over Robot Operating System (ROS) 2 with Data Distribution Service (DDS). Computation is distributed across three units: the Spot onboard computer SpotCore (base control and odometry), a NVIDIA Jetson (perception and arm control), and an external PC (mission coordination and WBC).

A. Hardware Platform

The Boston Dynamics Spot quadruped is commanded via its onboard ROS 2 driver through velocity commands $[v_x, \omega_z]$ and body posture commands. The Unitree Z1 is a six-DOF manipulator mounted on Spot's back, controlled at the torque level through a Joint Trajectory Controller (JTC) interface. Two RGBD cameras provide complementary coverage. An Orbbec Femto Bolt is mounted on Spot's body for wide-field patient detection at distances up to $3\,\mathrm{m}$. An Intel RealSense D435 is mounted at the Z1 EE; the arm's WBC look-at controller positions it as an active pan-tilt unit for close-range torso tracking and surface estimation. Spot's five built-in stereo cameras are positioned low on the body for ground-plane obstacle avoidance and are not suitable for YOLO skeleton detection.

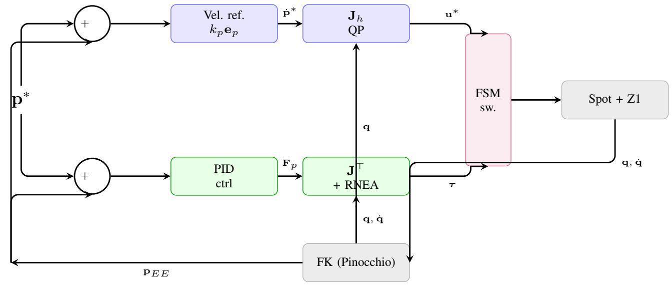

Fig. 2. Control loop diagram showing the two actuation paths: the WBC QP path for positioning and the Impedance Control path with FK feedback for compliant contact during ultrasound scanning.

B. Mission Management and System Integration

The mission is orchestrated by a 13-state FSM (Fig. 3). The robot first waits for Transform (TF) readiness, then searches for the patient through body rotation and dual-sensor confirmation. Once localised, it collects approach points, straightens its posture, and drives toward the target while the anticipatory scanner runs concurrently. After approach, it executes a posture-adaptive exposure body scan with integrated wound and burn detection, followed by an interactive review phase where the operator can click any grid point for re-inspection. The mission concludes with FAST ultrasound scanning under pre-planned body reconfiguration. Transitions are triggered by perception confidence (dashed) or geometric conditions (solid).

Fig. 4 shows the complete system architecture in six layers. The Orbbec and RealSense RGBD streams feed the Perception layer, whose outputs route to the FSM. The FSM enables four Planning modules: the Exposure Scanner (body scan grid generation and interactive review), the FAST Ultrasound planner, the Body Pose Optimizer, and the Web UI. The Control layer translates plans into actuation: the WBC QP publishes base velocity $[v_x, \omega_z]$ to Spot and arm goals to the IK Goal Multiplexer; the Impedance Controller manages compliant contact during FAST scanning, publishing joint torques $\boldsymbol{\tau}$ directly to the Z1. The Z1 FSM synchronises with the WBC coordinator: during the FAST cycle posture changes are applied before each target arm trajectory. An emergency stop immediately halts the base and returns the arm to a safe configuration.

Fig. 2 illustrates the two actuation paths.

C. Web-Based Operator Interface

Full autonomy in emergency assessment is neither feasible nor desirable: the operator must retain the ability to override or re-inspect regions of clinical interest. TERESA bridges

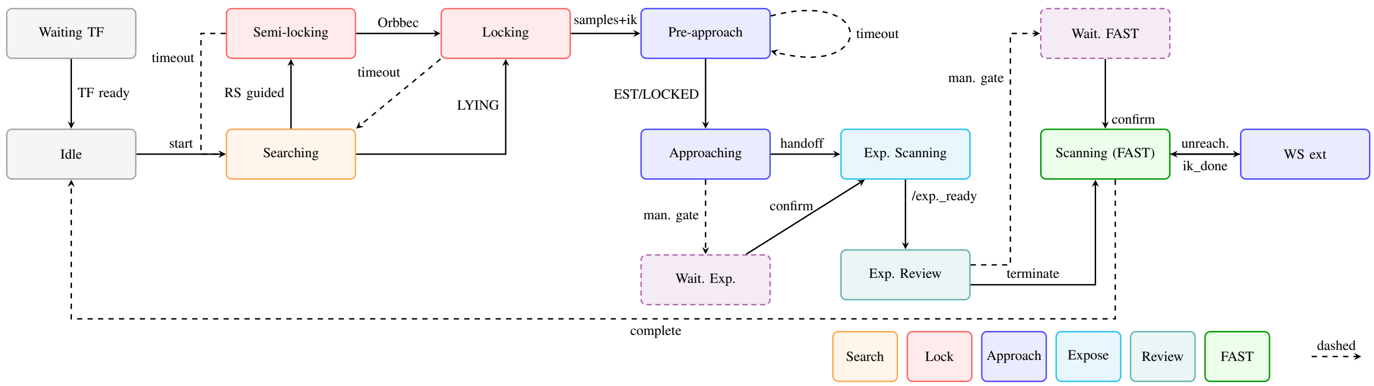

Fig. 3. Mission FSM with 13 states. The robot searches for the patient through body rotation and dual-sensor confirmation (SEARCHING → SEMI_LOCKING → LOCKING), approaches while running the anticipatory scanner (PRE_APPROACH → APPROACHING), performs a posture-adaptive exposure body scan with integrated wound and burn detection (EXPOSURE_SCANNING), enters an interactive review phase for operator-driven re-inspection (EXPOSURE_REVIEW), and executes FAST ultrasound scanning with pre-planned body reconfiguration (SCANNING). Transitions are triggered by perception confidence (dashed) or geometric conditions (solid).

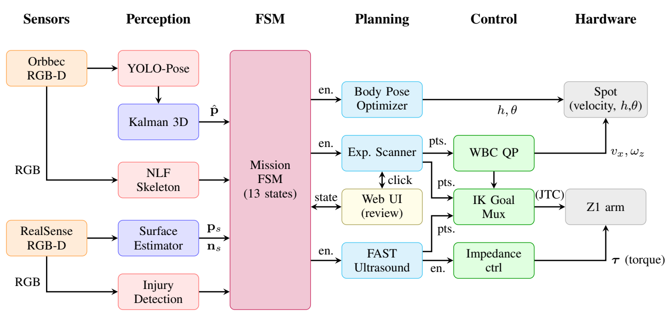

Fig. 4. System block diagram. The Perception layer processes dual-sensor RGBD streams via YOLO-pose, Kalman filtering, **NLF!** skeleton detection, surface estimation, and injury detection. The 13-state Mission FSM orchestrates four Planning modules: Exposure Scanner, FAST Ultrasound, Body Pose Optimizer, and the Web UI for interactive review. The Control layer executes real-time decisions through the WBC QP, IK Goal Multiplexer, and Impedance Controller. Dashed arrows denote operator-triggered or posture-optimisation paths.

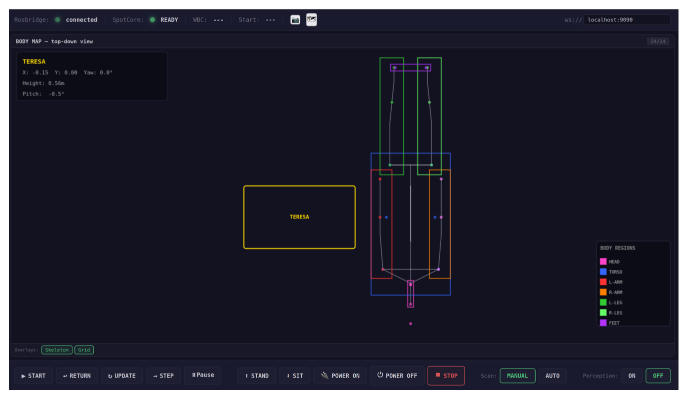

Fig. 5. Web-based operator interface during the exposure scanning phase. The Body Map (top-down view) shows the patient mannequin skeleton with colour-coded body regions, the exposure scan grid points, and the Spot robot position (yellow). The info panel (top-left) displays Spot's current posture parameters. During exposure review, the operator can click any grid marker to command the robot to re-inspect that body region.

this gap with a web interface that translates a single 2D click on the Body Map into a 3D whole-body repositioning command—posture optimisation, base navigation, and arm trajectory replay—without manual teleoperation.

The TERESA web interface provides dual-camera monitoring and robot control via a browser dashboard communicating over rosbridge WebSocket. During exposure review, the RealSense overlay renders the scan grid as clickable markers; each click triggers a full re-positioning sequence—posture optimisation, settle, and IK trajectory replay—within the same ROS 2 topic protocol used for autonomous operation. A manual/automatic scan gate toggle allows the operator to require explicit confirmation before advancing between mission phases. All web commands are mirrored on the keyboard controller for redundancy. Fig. 5 shows the interface during review.

V. Experimental Results

A. Experimental Setup

Experiments ran on the physical Spot+Z1 platform with an Orbbec Femto Bolt on Spot's body and a RealSense D435 at the EE, computing across SpotCore, an NVIDIA Jetson, and an external PC over ROS 2 Jazzy. Ten supine patient positions were defined ($2\,\mathrm{m}$–$4\,\mathrm{m}$, $\pm 45^\circ$ angular offset); each ran in both modes below, yielding twenty paired trials.

- **Baseline**: sequential stop-and-manipulate. Fixed three-pose scan after navigation, no concurrency.
- **TERESA**: anticipatory scanner overlapping navigation, adaptive grid, and QualityMonitor-modulated velocity.

Metrics: total mission time, idle time, scan overlap (fraction of body scan completed before handoff), EE positioning error at each FAST point, and lock time.

B. Search Phase: Whole-Body Coverage and Dual-Sensor Lock

We isolated the arm's contribution to search through an ablation experiment: TERESA (full system) versus a Spot-only baseline with the Z1 arm disabled and only the front-facing Orbbec active. Twenty trials were conducted at five body yaw angles ($0°$–$180°$, two repetitions each), measuring time to Orbbec posture confidence ≥ 0.70.

With the arm's seven scanning poses providing full $360°$ coverage from a single base position, TERESA required no base rotation: frontal patients were detected near-instantly, the rest during the scan, yielding a mean of $3.2(11)\,\mathrm{s}$. Without the arm, the fixed camera's $77°$ effective field of view forced five discrete base rotations ($\sim$$7\,\mathrm{s}$ each at $0.2\,\mathrm{rad\,s^{-1}}$, plus $2.0\,\mathrm{s}$ dwell), giving $14.8(23)\,\mathrm{s}$—a **5.0**$\times$ slowdown. A $10\,000$-trial Monte Carlo simulation confirmed this ratio across all patient orientations.

The dual-sensor architecture added robustness independently of speed: in three of the ten TERESA trials, the Orbbec alone failed to lock. These failures stemmed from the Orbbec's fixed horizontal viewpoint at $0.5\,\mathrm{m}$ above the ground, which is suboptimal for detecting a supine body whose visual features—torso, limbs, head—are primarily visible from above. The arm-mounted RealSense, positioned by the WBC look-at controller to observe the scene from an elevated angle, detected partial torso keypoints in these cases and guided the base toward the body, giving the Orbbec a second observation window at a more favourable orientation. Without this redundancy, these trials would have resumed the coarse search cycle, adding $\sim$$14\,\mathrm{s}$. The arm thus serves two complementary roles: its $360°$ coverage eliminates the base-rotation bottleneck, while its elevated viewpoint provides the sensor redundancy that prevents detection failures a single-camera system would incur.

C. End-to-End Mission Performance

Across all twenty end-to-end trials, TERESA reduced total mission time from $100.2(81)\,\mathrm{s}$ to $89.7(73)\,\mathrm{s}$ ($10.5\,\%$). The gain was concentrated in idle time: in the baseline, the body scan executed after the base stopped, adding $\sim$$10\,\mathrm{s}$ of idle time; TERESA absorbed this into the approach, completing $\sim$$80\,\%$ of the scan before reaching the handoff point. The adaptive scan grid accelerated this further: in the majority of trials, torso keypoints were already resolved before approach, and the grid automatically selected the minimal two-pose pattern rather than the full three-pose sweep. Pre-planned body reconfiguration reduced probe positioning error from $44.7(52)\,\mathrm{mm}$ to $29.8(41)\,\mathrm{mm}$, with the largest improvement at lateral FAST targets. The exposure scan dominated the mission budget at $\sim$$40\,\mathrm{s}$ in both configurations, limiting the overall speedup. In manual operation, operators could skip unreachable grid points, reducing total time to $\sim$$70\,\mathrm{s}$ at the cost of incomplete coverage.

D. Whole-Body Reconfiguration

The body posture optimizer described in Section III-C proved essential for reaching distant anatomical targets on a full-scale body. We compared four configurations of increasing capability. The fixed nominal posture (height $0\,\mathrm{m}$, pitch $0°$, no base displacement), representing Spot's default standing pose, reached only $\sim$$40\,\%$ of exposure grid points and $\sim$$50\,\%$ of FAST targets. A heuristic posture—moderately lowered ($\sim$$-0.15\,\mathrm{m}$) with a slight forward pitch ($\sim$$5°$)—improved these to $\sim$$55\,\%$ and $\sim$$65\,\%$, but could not adapt to individual point locations. The 2D optimiser, selecting the best (h, φ) per point via grid search with IK-driven retry, raised completion to $\sim$$70\,\%$ (exposure) and $\sim$$85\,\%$ (FAST). Finally, the 3D optimiser added longitudinal base walking dy with the regularised cost $J = \|\mathbf{p} - \mathbf{p}_{\mathrm{sweet}}\| + \lambda\,|dy|$ ($\lambda = 0.50$), achieving $\sim$$85\,\%$ and $100\,\%$ respectively. The penalty encouraged height and pitch adjustments over walking—reducing energy consumption and producing smoother motion—while still allowing displacement when necessary. The residual $15\,\%$ of exposure points were at the furthest extremities (head, feet), beyond the arm's maximum reach even with full base repositioning; these were skipped gracefully. Each tier contributed measurably to mission completeness, with the movement penalty ensuring that base walking is used only when strictly needed.

E. Exposure Body Scanning and Injury Detection

The exposure pipeline was evaluated with XX trials on supine patients, generating XX $\pm$ XX scan points. Per-point dwell of $2\,\mathrm{s}$ yielded scan durations of XX $\pm$ XX s. The injury detection pipeline achieved a wound recall of XX % and burn classification accuracy of XX %, with a false positive rate of XX per scan on average. The 3D localisation error of detected injuries was XX $\pm$ XX mm.

F. Discussion

The results demonstrate that treating body posture as an active perceptual resource yields measurable improvements in both mission speed and completeness. The arm's $360°$ coverage eliminates the base-rotation bottleneck during search, while body reconfiguration lifts completion from $\sim$$40\,\%$ to near-$100\,\%$. The dual-sensor architecture prevents detection failures that a single-camera system would incur. Several limitations remain: the system assumes a single stationary patient on flat terrain; ultrasound image quality and clinical scanning pressure were not evaluated; and the injury detection pipeline relies on pre-trained models whose performance may vary across wound types and skin tones. These aspects are the subject of ongoing work.

VI. Conclusion

This paper presented TERESA, an integrated system for autonomous emergency assessment on a legged mobile manipulator. The core contribution is a whole-body active perception framework that treats body posture—height, pitch, and yaw—as a first-class perceptual resource, closing the perception-action loop at the whole-body level. Experiments on the physical Spot+Z1 platform demonstrated that the arm's $360°$ coverage accelerates patient search by a factor of five compared to a fixed-camera baseline; that the dual-sensor lock prevented detection failures in three out of ten trials, while the adaptive anticipatory scanner completed 78% of the body scan concurrently with approach navigation, eliminating idle time. The adaptive scan grid automatically selected the minimal two-pose pattern in the majority of trials, with the three-pose comprehensive pattern reserved for cases with incomplete keypoint detection; and that body reconfiguration lifts anatomical target completion from $\sim 40\,\%$ to near-$100\,\%$ across FAST and exposure scanning. The exposure pipeline supports supine postures with integrated wound and burn detection and an interactive web-based review interface. Future work will extend TERESA to multi-patient scenarios and clinical ultrasound image acquisition with impedance-controlled probe contact. A further direction is the validation of the FAST ultrasound and impedance control pipeline on human subjects, in accordance with ethical approval, comparing the acquired images against those obtained by specialist clinicians to assess diagnostic reliability. Beyond the Exposure and FAST components demonstrated here, the TERESA platform could be extended to cover the full ABCDE protocol—Airway, Breathing, Circulation, Disability, and Exposure— providing a complete autonomous primary survey for emergency triage.

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